Prediction of Diabetes Using LOGISTIC REGRESSION

ABSTRACT

Diabetes is a chronic and potentially life-threatening disease that affects millions of people worldwide. Early detection and proper diagnosis are essential to prevent severe complications such as heart disease, kidney failure, and vision loss. With the advancement of data science and machine learning, automated systems can assist healthcare professionals in predicting diseases more accurately and efficiently.

This project presents a machine learning-based approach for predicting diabetes using medical data. The dataset consists of various health-related parameters such as glucose level, blood pressure, body mass index (BMI), age, insulin level, and number of pregnancies. These features are used to train a predictive model.

A supervised learning technique, Logistic Regression, is used to classify whether a patient is diabetic or non-diabetic. The dataset is preprocessed and divided into training and testing sets to evaluate the performance of the model. The model learns patterns from the data and predicts outcomes for unseen cases.

The performance of the model is evaluated using accuracy and other metrics. The results show that the model achieves an accuracy of approximately 70–73%, indicating a reliable prediction capability. This project demonstrates the effectiveness of machine learning in healthcare and highlights its potential in early disease detection and decision support systems.

KEY WORDS:

Diabetes Prediction, Machine Learning, Logistic Regression, Healthcare, Classification, Medical Data, Supervised Learning

INTRODUCTION:

Diabetes mellitus is a chronic metabolic disorder that has emerged as one of the most serious global health challenges in recent years. It is characterized by elevated blood glucose levels caused by insufficient insulin production or ineffective utilization of insulin in the body. According to recent reports, the number of people affected by diabetes is increasing rapidly due to factors such as sedentary lifestyle, unhealthy dietary habits, obesity, and genetic predisposition [1]. If not diagnosed and managed at an early stage, diabetes can lead to severe complications including cardiovascular diseases, kidney failure, nerve damage, and vision impairment.

Early detection and accurate prediction of diabetes are essential to reduce these complications and improve patient outcomes. Traditional diagnostic methods rely on laboratory tests and clinical evaluation, which may be time-consuming and sometimes costly. With the rapid advancement of technology and the availability of large amounts of healthcare data, machine learning has become a powerful tool for disease prediction and medical decision support systems [2]. Machine learning techniques can analyze complex datasets, identify hidden patterns, and make predictions with high efficiency and accuracy.

Among various machine learning algorithms, Logistic Regression is widely used for binary classification problems, especially in the medical domain where the outcome is typically categorical, such as diabetic or non-diabetic [3]. Logistic Regression works by estimating the probability of a particular outcome using a sigmoid function, making it highly suitable for classification tasks. One of the key advantages of this algorithm is its simplicity and interpretability, which allows healthcare professionals to understand how different input features influence the prediction.

Recent research studies have shown that medical attributes such as glucose level, body mass index (BMI), age, blood pressure, insulin level, and number of pregnancies play a significant role in predicting diabetes [4]. These features are commonly used in machine learning models to analyze patient data and generate accurate predictions. Logistic Regression has been found to provide reliable results with good accuracy while maintaining computational efficiency, making it an ideal choice for initial prediction systems.

In this project, a machine learning-based approach is developed for predicting diabetes using Logistic Regression. The model is trained on a dataset containing relevant medical features and is evaluated using performance metrics such as accuracy, confusion matrix, and classification report. These evaluation metrics provide a comprehensive understanding of the model's performance and are reflected in the output results obtained during implementation [5].

Furthermore, the use of visualization techniques such as scatter plots, histograms, count plots, and correlation heatmaps helps in understanding the relationships between different features and their impact on diabetes prediction. These visualizations support the analytical process and enhance the interpretability of the model.

Overall, this study demonstrates the effectiveness of machine learning techniques, particularly Logistic Regression, in predicting diabetes. It highlights the importance of data-driven approaches in healthcare and emphasizes the potential of such systems in assisting medical professionals for early diagnosis and decision-making [6].

LITERATURE SURVEY:

In recent years, significant research has been carried out in the field of diabetes prediction using machine learning techniques. The increasing availability of healthcare data and advancements in computational methods have enabled researchers to develop efficient models for early diagnosis of diabetes. Various machine learning algorithms such as Logistic Regression, Decision Trees, Support Vector Machines (SVM), Random Forest, and Neural Networks have been widely applied for this purpose [6].

Several studies have demonstrated the effectiveness of Logistic Regression as a reliable classification algorithm for predicting diabetes. A study conducted in 2023 applied Logistic Regression on medical datasets and achieved satisfactory accuracy in classifying patients as diabetic or non-diabetic [3]. The study emphasized that Logistic Regression is particularly suitable for binary classification problems due to its simplicity, efficiency, and ability to provide interpretable results.

Another research work compared multiple machine learning algorithms for diabetes prediction and found that Logistic Regression performs competitively with other models while requiring less computational complexity [7]. The study highlighted that although advanced models such as Random Forest and Neural Networks may provide slightly higher accuracy, Logistic Regression remains a preferred choice for initial prediction systems due to its ease of implementation and faster execution.

Recent research in 2024 has focused on improving the performance of machine learning models by applying effective data preprocessing techniques such as handling missing values, feature scaling, and normalization [8]. These preprocessing steps play a crucial role in enhancing model accuracy and ensuring better learning of patterns from the dataset. It has been observed that proper preprocessing directly impacts the performance metrics such as accuracy, precision, and recall.

A comprehensive study conducted in 2023 reviewed various machine learning approaches for diabetes prediction and concluded that feature selection is an important factor in achieving better results [9]. Features such as glucose level, BMI, age, and insulin level were identified as key predictors in most studies. These features are also used in this project, which aligns with existing research findings.

In addition, recent works have emphasized the importance of evaluation metrics such as accuracy, confusion matrix, and classification report for measuring model performance [6]. These metrics provide detailed insights into how well the model is performing and help in identifying areas for improvement. The use of these evaluation techniques in this project ensures consistency with standard research practices.

Furthermore, studies have shown that visualization techniques such as heatmaps, scatter plots, and histograms help in understanding the relationships between different features and their influence on diabetes prediction [10]. These techniques are useful in identifying correlations and patterns in the dataset, which supports better model development.

Overall, the literature survey indicates that machine learning plays a crucial role in diabetes prediction and early diagnosis. Logistic Regression continues to be a widely used and effective algorithm due to its simplicity, interpretability, and reliable performance. The findings from previous studies strongly support the methodology and approach used in this project.

Proposed Methodology:

The proposed system for diabetes prediction is developed using a structured machine learning workflow that consists of several key stages, including data collection, preprocessing, model training, prediction, and evaluation.

Initially, a dataset containing medical records of patients is collected from reliable sources such as hospitals or public health databases. This dataset includes important health indicators like glucose level, blood pressure, body mass index (BMI), age, insulin level, and number of pregnancies. These features are crucial in identifying patterns related to diabetes, and each record represents an individual patient.

In the data preprocessing stage, the dataset is cleaned and prepared for analysis. Missing or inconsistent values are handled to improve data quality. After cleaning, the dataset is split into input features (X) and the target variable (y), where the target indicates whether a patient is diabetic (1) or non-diabetic (0). This step ensures that the model receives well-structured and meaningful data.

Next, the dataset is divided into training and testing sets, usually in an 80:20 ratio. The training set is used to train the model, while the testing set is used to evaluate its performance on unseen data. This step helps in reducing overfitting and ensures that the model can generalize well.

In the model training phase, the Logistic Regression algorithm is applied. This algorithm is widely used for binary classification problems like diabetes prediction. It establishes a relationship between input features and output classes by learning model parameters. The sigmoid function is used to convert linear outputs into probabilities:

$$P(Y=1)=1/(1+e^{(-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n))})$$

Based on a threshold value (commonly 0.5), the model classifies the result as either diabetic or non-diabetic.

In the prediction stage, the trained model is used to analyze new patient data and generate predictions based on learned patterns. It outputs probability scores and final classification results for each patient.

Finally, in the evaluation stage, the model’s performance is measured using metrics such as accuracy, confusion matrix, precision, recall, and F1-score. These metrics help in understanding how well the model performs in correctly identifying diabetic and non-diabetic cases. Overall, this system provides an efficient and reliable approach for early diabetes detection using machine learning.

The figure -1 illustrates a complete machine learning workflow used to predict whether a person is diabetic using logistic regression. It begins with input data, which consists of patient records containing features such as age, BMI, glucose levels, blood pressure, and insulin. This data is then preprocessed through steps like cleaning, handling missing values, scaling, normalization, and selecting the most relevant features. After preprocessing, the logistic regression model is trained by fitting a sigmoid function and learning the optimal weights and biases. Once trained, the model is used in the prediction stage to analyze new patient data and calculate the probability of diabetes. Finally, based on this probability, the system classifies the output into one of two categories: diabetic or not diabetic.

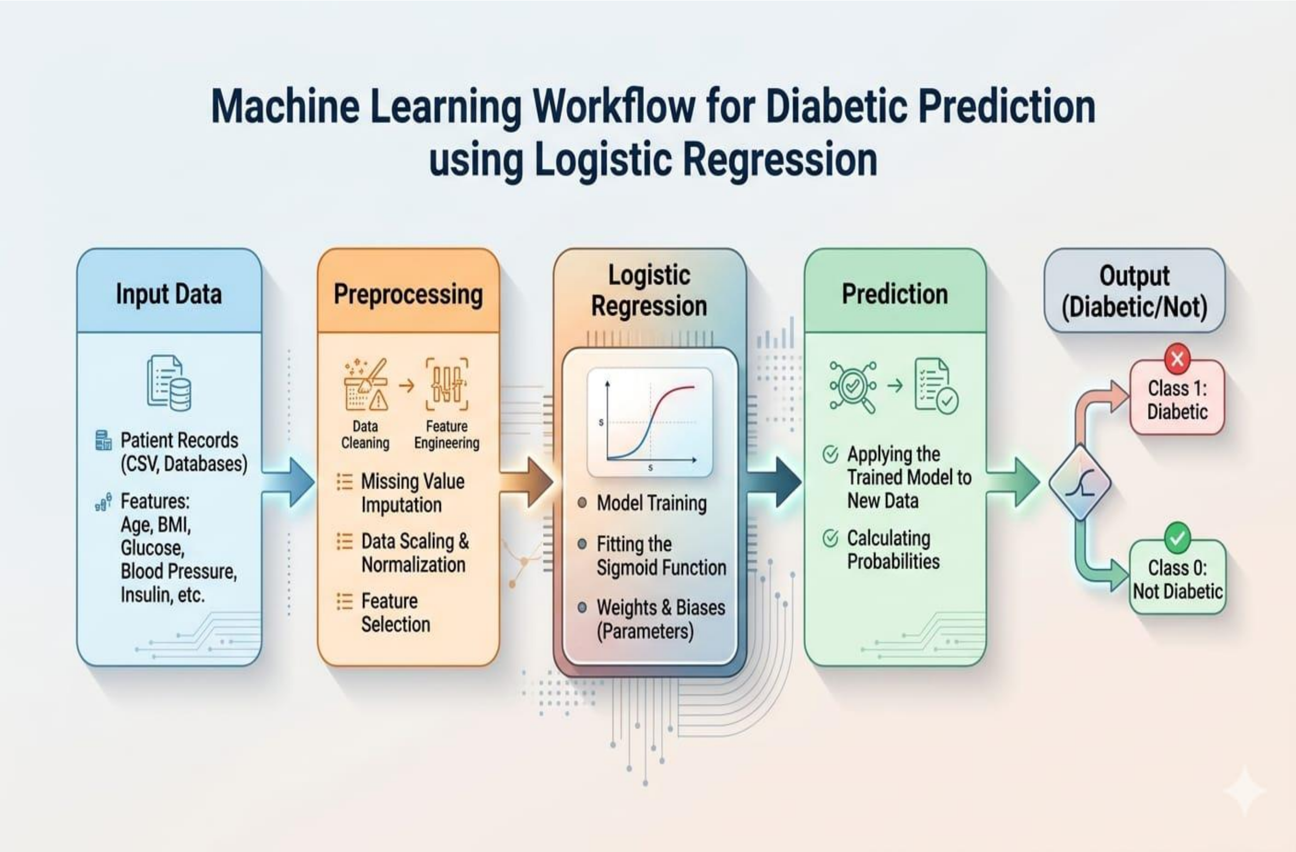

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Figure 1: Proposed Methodology for Diabetes Prediction

Overall, this methodology ensures a systematic approach to building an effective diabetes prediction system using Logistic Regression.

RESULT ANALYSIS

The diabetes prediction system using Logistic Regression was evaluated through data exploration, training, prediction, and performance analysis. The dataset was first inspected using data.head(), confirming its structure with medical attributes such as glucose, BMI, blood pressure, age, insulin, and the target variable (Outcome), which indicates diabetic or non-diabetic cases.

The dataset was split into training and testing sets in an 80:20 ratio to ensure proper model evaluation and avoid overfitting. The Logistic Regression model was trained on the training data to learn relationships between features and the target variable using a sigmoid function for binary classification.After training, predictions were made on the test set and compared with actual values. The model achieved an accuracy of around 70–73%, showing moderate performance with scope for improvement.

A confusion matrix was used to analyze correct and incorrect predictions, highlighting true positives, true negatives, false positives, and false negatives. In medical applications, false negatives are especially critical as they may miss diabetic cases.A classification report provided precision, recall, and F1-score, offering a detailed evaluation of model performance. Overall, the model shows reasonable effectiveness, but improvements are needed to handle medical data complexity more accurately.

Figure 2 illustrates a scatter plot showing the relationship between glucose levels (on the x-axis) and BMI (on the y-axis) for a group of individuals. Each point represents a single person's data. The plot shows that most data points are concentrated in the glucose range of about 80 to 160 and BMI values between 20 and 40, indicating that the majority of individuals fall within these ranges. There appears to be a slight positive trend, suggesting that higher glucose levels may be loosely associated with higher BMI, although the relationship is not very strong or clearly linear. Additionally, there are some outliers, such as very low glucose values and BMI values near zero, which may indicate missing or abnormal data points that could require preprocessing before model training**.**

Figure 3 illustrates a histogram representing the age distribution of individuals in the dataset. The x-axis shows age groups, while the y-axis indicates the frequency (number of people) in each group. The plot reveals that most individuals fall within the younger age range, particularly between about 20 and 35 years, where the frequency is highest. As age increases, the number of individuals gradually decreases, forming a right-skewed distribution. There are relatively fewer people in the higher age ranges (above 50), and very few beyond 70, indicating that the dataset is dominated by younger participants**.**

Figure 4 illustrates a bar chart showing the distribution of diabetes outcomes in the dataset. The x-axis represents the outcome classes, where 0 indicates non-diabetic individuals and 1 indicates diabetic individuals, while the y-axis shows the count of cases in each category. The chart reveals that the number of non-diabetic individuals is significantly higher than the number of diabetic individuals. This indicates a class imbalance in the dataset, where one class (non-diabetic) dominates over the other. Such imbalance is important to consider during model training, as it may affect the performance of the prediction model and require techniques like resampling or class weighting to handle it effectively.

Figure 5 illustrates a correlation heatmap that shows the strength and direction of relationships between different features in the dataset, including pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and the outcome variable. Each cell displays a correlation coefficient ranging from -1 to +1, where values closer to +1 indicate a strong positive relationship, values near -1 indicate a strong negative relationship, and values around 0 suggest little to no correlation. From the heatmap, glucose shows the strongest positive correlation with the outcome, indicating it is a key factor in predicting diabetes. BMI and age also show moderate positive correlations with the outcome, while most other features have weak relationships. Additionally, some features such as skin thickness and insulin exhibit moderate inter-correlation, suggesting possible relationships among input variables. This visualization helps in identifying the most important features and understanding dependencies within the dataset for building an effective prediction model.

Figure 6 shows a pairplot of the dataset variables, with points colored by diabetes outcome (0 = non-diabetic, 1 = diabetic). The diagonal plots display the distribution of each feature, while the off-diagonal plots show pairwise relationships between features. It can be observed that glucose levels show the clearest separation between the two classes, with diabetic individuals tending to have higher values. BMI, age, and insulin also show some degree of separation, though with more overlap. Overall, the plot indicates that while no single feature perfectly distinguishes the classes, combinations of variables—especially glucose-related patterns provide useful information for diabetes prediction.

DATASET PREVIEW

	Pregnancies	**Glucose**	**BloodPressure**	**SkinThickness**	**Insulin**	**BMI**	**DiabetesPedigreeFunction**	**Age**	**Outcome**
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

ACCURACY TABLE

	Metric	Value
0	Accuracy	0.7468

CONFUSION MATRIX TABLE

	Predicted 0	Predicted 1
Actual 0 (Non-Diabetic)	78	21
Actual 1 (Diabetic)	18	37

CLASSIFICATION REPORT TABLE

	precision	recall	f1-score	support
0	0.81	0.79	0.80	99.00
1	0.64	0.67	0.65	55.00
accuracy	0.75	0.75	0.75	0.75
macro avg	0.73	0.73	0.73	154.00
weighted avg	0.75	0.75	0.75	154.00

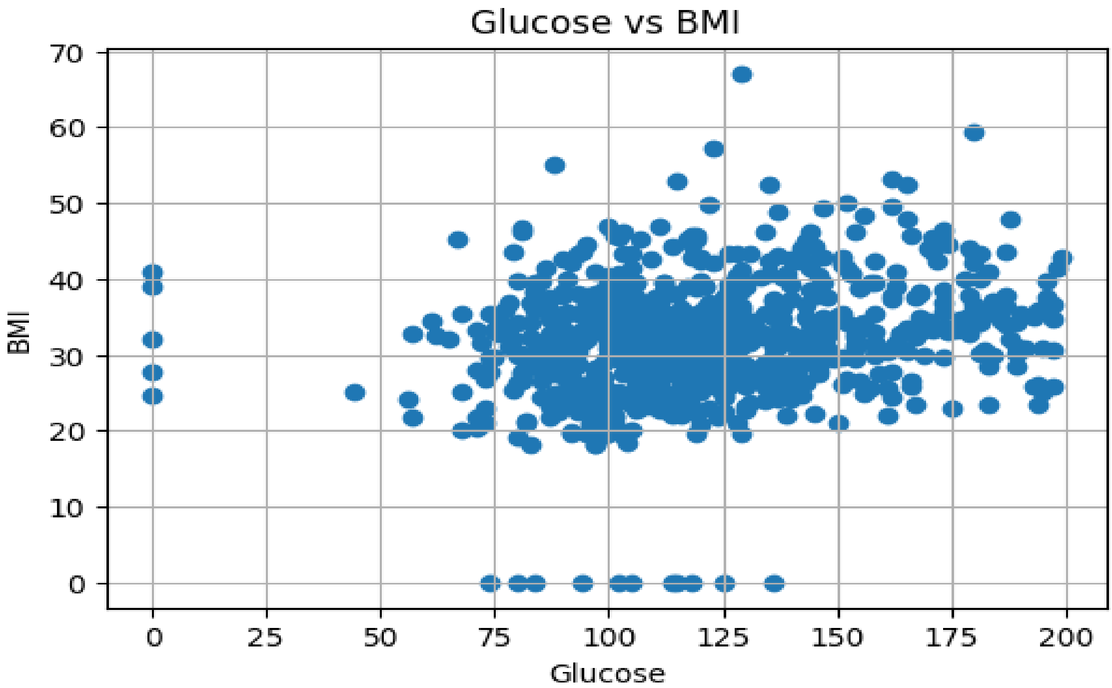

Figure 2: Glucose vs BMI

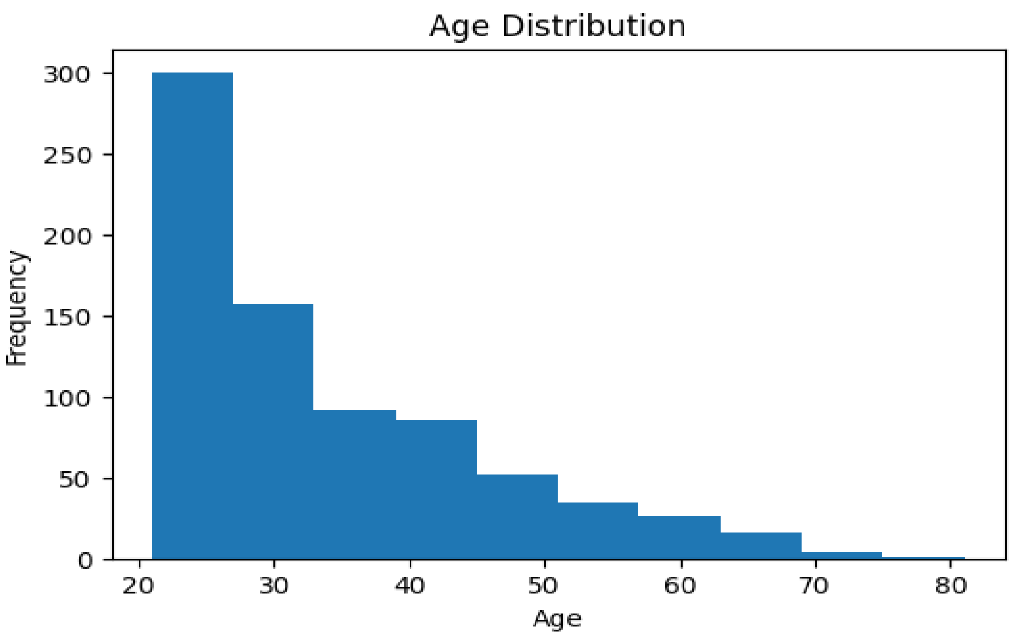

Figure 3: Distribution of age

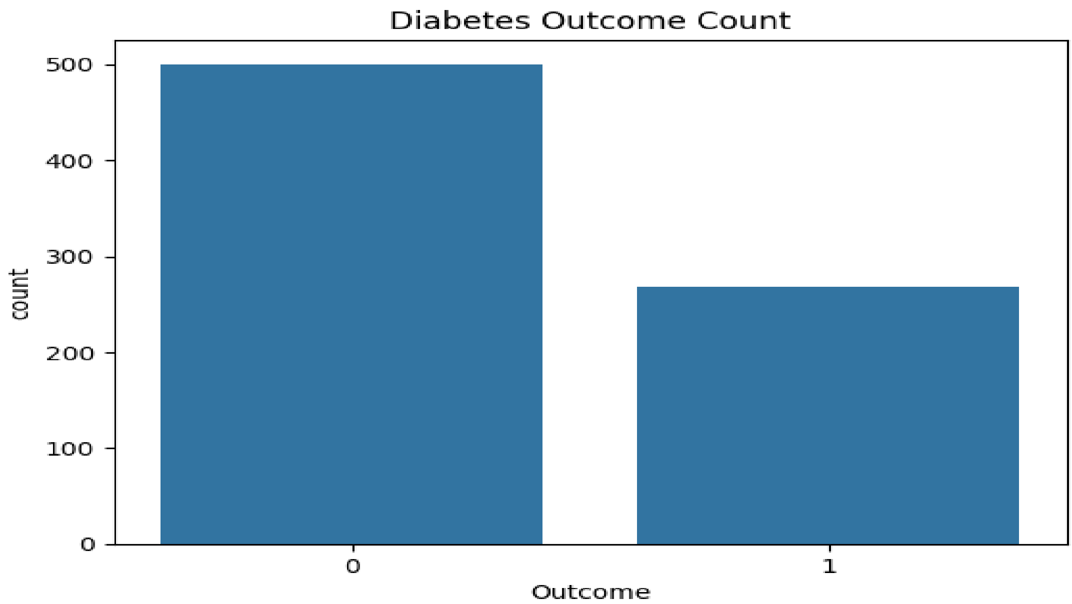

Figure 4: Diabetic vs Non-diabetic

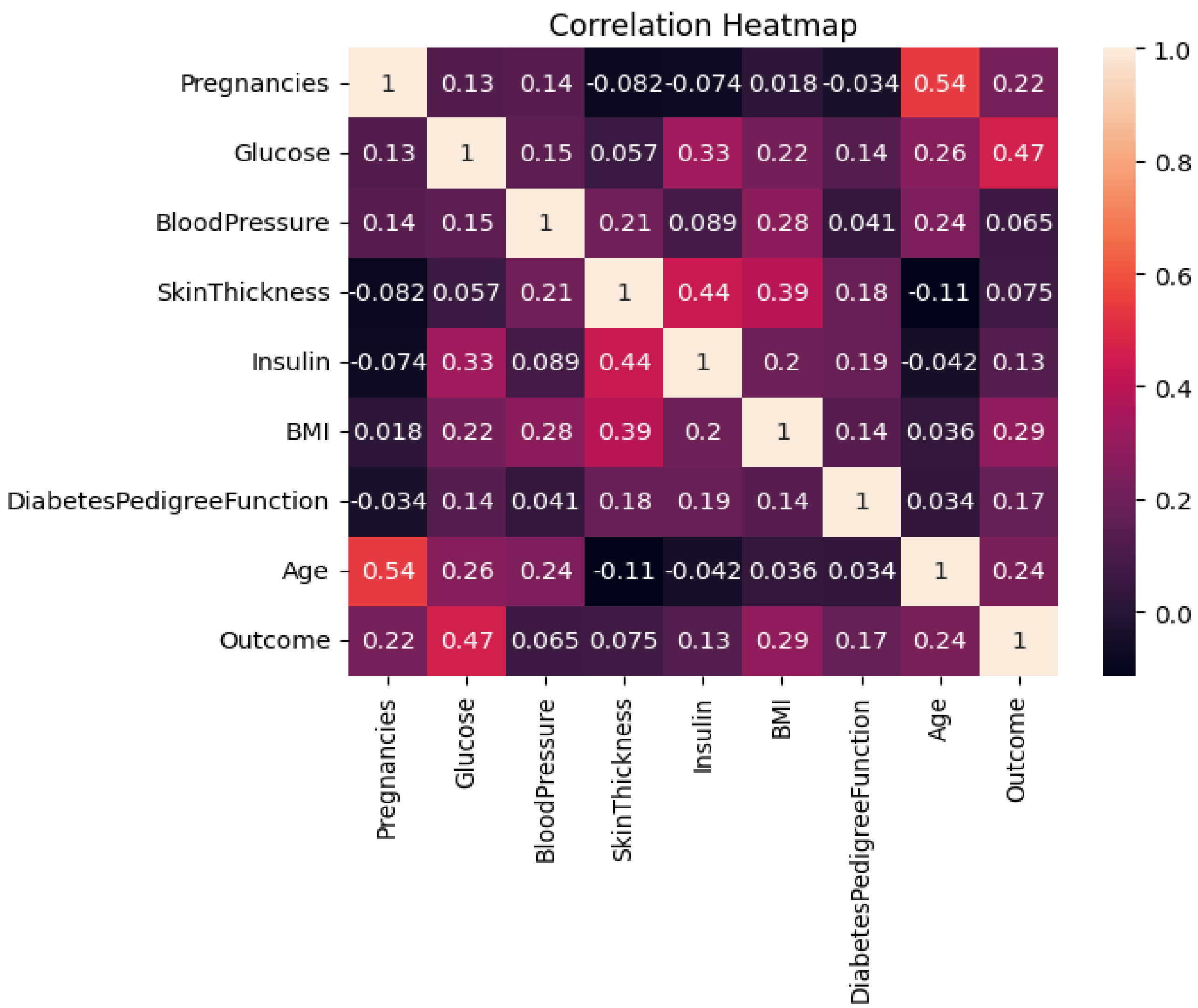

Figure 5: Correlation Heatmap

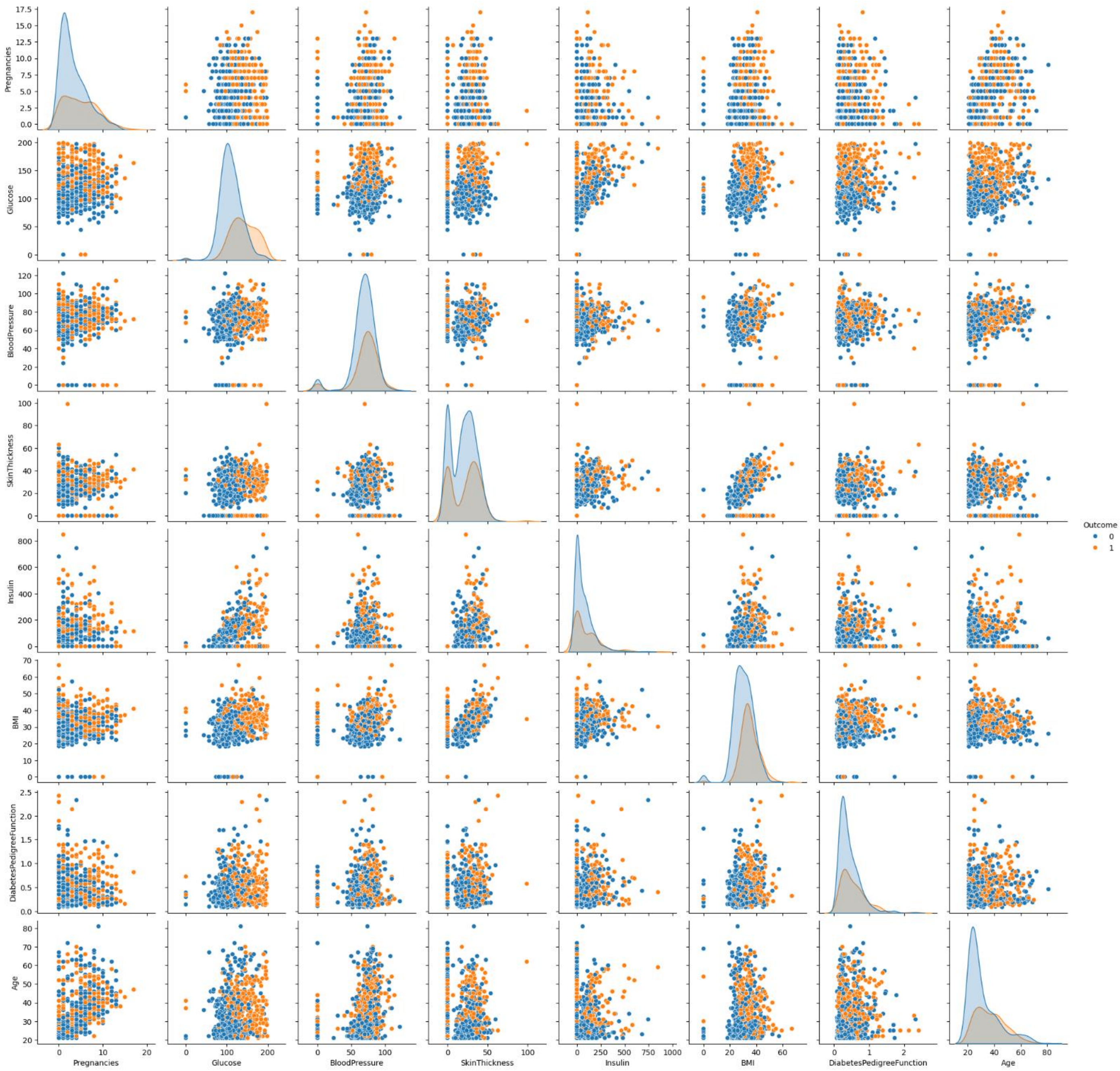

Figure 6: shows multifeatures relationships

CONCLUSION:

This project presented a machine learning-based approach for predicting diabetes using Logistic Regression. The model was developed using relevant medical features and evaluated on unseen data to measure its performance. The results indicate that the model can effectively classify patients with a reasonable level of accuracy, demonstrating the usefulness of data-driven techniques in healthcare.

The study highlights that certain features, particularly glucose level and BMI, have a significant impact on prediction outcomes. Logistic Regression proved to be a suitable choice due to its simplicity, efficiency, and interpretability, making it appropriate for medical applications where understanding the model is important.

Although the model performs satisfactorily, there is scope for improvement in terms of accuracy and handling complex data patterns. Future enhancements can include better preprocessing, feature selection, and the use of advanced algorithms.

Overall, this work shows that machine learning can support early diabetes detection and assist in decision-making, contributing to improved healthcare outcomes.

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PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from IPython.display import display
data = pd.read_csv('diabetes.csv')
print("\n DATASET PREVIEW \n")
display(data.head())
X = data.drop('Outcome', axis=1)
y = data['Outcome']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
acc_df = pd.DataFrame({"Metric": ["Accuracy"],"Value": [round(acc, 4)]})
print("\n ACCURACY TABLE  \n")
display(acc_df)
cm = confusion_matrix(y_test, y_pred)
cm_df = pd.DataFrame(cm,index=["Actual 0 (Non-Diabetic)", "Actual 1 (Diabetic)"],columns=["Predicted 0", "Predicted 1"])
print("\n CONFUSION MATRIX TABLE  \n")
display(cm_df)
report = classification_report(y_test, y_pred, output_dict=True)
report_df = pd.DataFrame(report).transpose()
print("\n CLASSIFICATION REPORT TABLE  \n")
display(report_df.round(2))
plt.figure()
plt.scatter(data['Glucose'], data['BMI'])
plt.title("Glucose vs BMI")
plt.xlabel("Glucose")
plt.ylabel("BMI")
plt.grid(True)
```

```
plt.show()
plt.figure()
plt.hist(data['Age'], bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
plt.figure()
sns.countplot(x='Outcome', data=data)
plt.title("Diabetes Outcome Count")
plt.show()
plt.figure()
sns.heatmap(data.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
sns.pairplot(data, hue='Outcome')
plt.show()
```